

Chapter 6: Graphical Representation of Causal Effects

Applied Use Case: Did the Retention Offer Actually Reduce Churn?

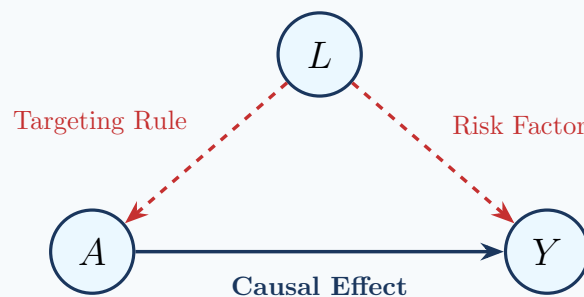
1 Introduction to Causal Directed Acyclic Graphs (DAGs)

Reference: *What If* (Hernán & Robins), Section 6.1, pp. 69–71.

In observational business analytics, identifying cause-and-effect relationships requires explicit, transparent assumptions. **Causal Directed Acyclic Graphs (DAGs)** provide a mathematical and visual language to map these assumptions, detect confounding pathways, and avoid selection bias traps.

- **Nodes (Vertices):** Represent random variables in our CRM ecosystem (e.g., Treatment A , Outcome Y , Covariates L).
- **Directed Edges (Arrows):** Represent direct causal relationships ($A \rightarrow Y$). The presence of an arrow denotes a potential causal effect; its absence represents a strict assumption of no direct causal effect.
- **Acyclic:** Time flows strictly forward. No feedback loops are allowed at a single point in time (e.g., $A \rightarrow Y \rightarrow A$ within the same timeframe is forbidden).

Figure 6.1: Basic Confounding DAG ($A \leftarrow L \rightarrow Y$)



Variable Legend & CRM Context:

- **A (Treatment):** Customer received retention discount ($A = 1$) vs. standard pricing ($A = 0$).
- **Y (Outcome):** Customer churned ($Y = 1$) vs. remained active ($Y = 0$).
- **L (Confounder):** Prior complaint history / drop in platform usage (baseline risk).

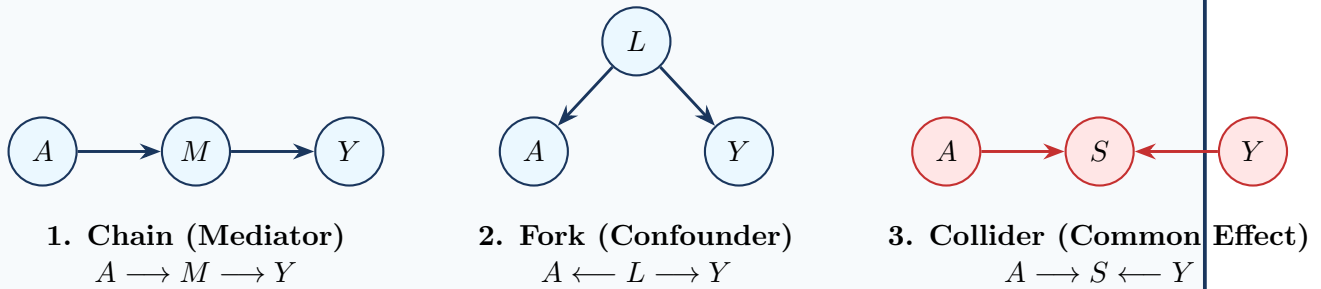
Key Structural Insight: The red dashed path $A \leftarrow L \rightarrow Y$ is an open **backdoor path**. Because dissatisfied customers ($L = 1$) are both more likely to receive offers ($A = 1$) and more likely to churn ($Y = 1$), naive associational analysis erroneously concludes that offers increase churn!

2 The Three Fundamental Structural Building Blocks

Reference: *What If*, Section 6.3, pp. 73–75; Technical Point 6.1, p. 77.

All complex causal DAGs are constructed from three elementary structural configurations. Understanding how association flows through these three structures is the foundation of causal inference.

Figure 6.2: The Three Structural Building Blocks of Causal Flow



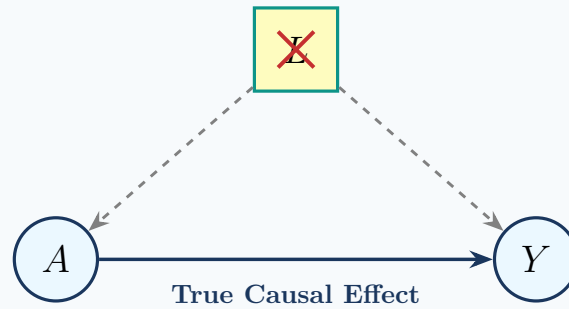
1. **Chain (Causal Path / Mediator M):** *Example:* Offer (A) $\rightarrow$ App Portal Log-ins (M) $\rightarrow$ Churn (Y).
 - **Status:** Natively **OPEN**. Transmits true causal effect from A to Y .
 - **Action:** Controlling for M **BLOCKS** the indirect causal path, leaving only direct effect.
2. **Fork (Spurious Path / Confounder L):** *Example:* Complaint History (L) affects both Offer Assignment (A) and Churn (Y).
 - **Status:** Natively **OPEN** (Backdoor Path). Induces non-causal association.
 - **Action:** Conditioning on L **BLOCKS** the backdoor path, eliminating confounding bias.
3. **Collider (Common Effect S):** *Example:* Offer (A) $\rightarrow$ Survey Response (S) $\leftarrow$ Intended Churn (Y).
 - **Status:** Natively **CLOSED / BLOCKED**. Does NOT induce confounding on its own.
 - **Action:** Conditioning on S (e.g., analyzing only survey respondents $S = 1$) **OPENS** the path, inducing **Selection Bias**!

3 Confounding Adjustment via the Backdoor Criterion

Reference: *What If*, Section 6.4, pp. 75–78; Technical Point 6.1.

To recover the true Average Causal Effect (ACE) from observational CRM data, we must identify a set of measured covariates L that satisfies the **Backdoor Criterion**:

1. L blocks all backdoor paths between Treatment A and Outcome Y .
2. L contains no descendants of Treatment A (to avoid blocking causal mediators or opening colliders).

Figure 6.3: Blocking Backdoor Paths by Conditioning on L **Mathematical Consequence of Adjustment:**

By conditioning on baseline confounder L (represented by the square box $\square$ and cross-out $\otimes$), the backdoor path $A \leftarrow L \rightarrow Y$ is blocked. Conditional Exchangeability is restored:

$$Y^a \perp\!\!\!\perp A \mid L \quad \text{for } a \in \{0, 1\}$$

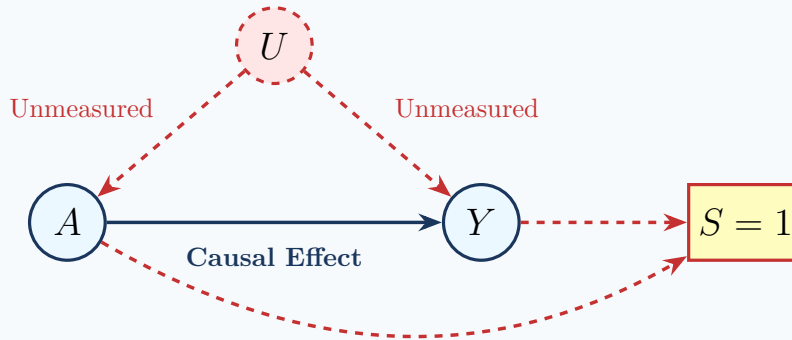
Methods such as **Standardization** or **Inverse Probability Weighting (IPW)** leverage this conditional exchangeability to calculate unbiased causal risk differences and risk ratios.

4 Unmeasured Confounding and Selection Bias

Reference: *What If*, Section 6.5, pp. 78–82; Figures 6.3–6.6.

Real-world observational CRM studies frequently suffer from two distinct structural threats to validity: **Unmeasured Confounding** and **Selection Bias**.

Figure 6.4: Structural Threats — Unmeasured Confounding (U) & Selection Bias (S)



Detailed Structural Threat Analysis:

1. Unmeasured Confounding (U):

- U represents unobserved customer attributes (e.g., unrecorded competitor price drops, general dissatisfaction).
- **Impact:** Because U is unmeasured, the backdoor path $A \leftarrow U \rightarrow Y$ **cannot be blocked**. Conditional exchangeability ($Y^a \perp\!\!\!\perp A \mid L$) is violated.
- **Takeaway:** No statistical algorithm (XGBoost, IPW, Regression) can eliminate bias caused by unmeasured confounders U .

2. Selection Bias ($S = 1$):

- $S = 1$ represents restricting analysis to customers who completed a post-campaign feedback survey.
- **Impact:** Both Offer (A) and Churn Intention (Y) influence Survey Completion (S). S is a **collider**.
- **Takeaway:** Restricting analysis to $S = 1$ (conditioning on a collider) **OPENS** a spurious non-causal path between A and Y , creating artificial correlation where none exists.

5 Summary: Practical Rules for Data Science Teams

Reference: *What If*, Section 6.6, pp. 82–84.

Executive Guidelines for Causal DAG Application

1. **Draw the DAG Before Modeling:** Map all domain hypotheses before building propensity scores or regression models.
2. **Adjust for Pre-Treatment Confounders (L):** Block all backdoor paths by conditioning on variables that affect both A and Y .
3. **Never Adjust for Post-Treatment Mediators (M):** Controlling for M blocks the very mechanism through which treatment works.
4. **Never Condition on Colliders (S):** Avoid filtering datasets by post-treatment selection variables (e.g., survey completion, app activity after offer).
5. **Acknowledge Unmeasured Confounders (U):** Use sensitivity analysis (e.g., E-values) when key confounders cannot be recorded in CRM.